Assignment 2

Deadline: Monday, 26 October 2026, 18:00

Introduction

In this assignment, you will investigate conditional independence statements encoded by Bayesian-network structures.

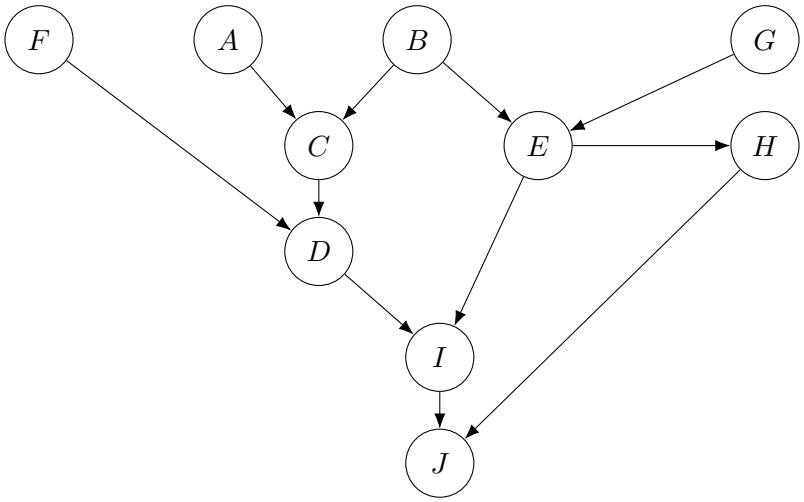
- Submit a PDF slide deck with up to two slides.
- "Expected content on the solution slide" gives instructions on what to put on your slides.

Do not forget to:

- indicate which parts of the assignment you have completed here
- select an interview time slot here;
- provide feedback here.

1 (Conditional) independence in Bayesian networks (4 points)

Consider the following Bayesian network:



Determine whether the following independence statements are implied by the graph:

(a) $I \perp I$,
(b) $B \perp J$,
(c) $F \perp G$,
(d) $C \perp E$,
(e) $A \perp H \mid J$,
(f) $C \perp E \mid B$,
(g) $G \perp F \mid I, C$,
(h) $J \perp A \mid C, I$

Hint. We have not formally covered the graphical criterion for answering all such independence questions in the lecture. Look up the concept of *d-separation* for Bayesian networks[1]. You may also find the *Bayes-ball algorithm* useful.

Expected content on the solution slide: Report your final answer to all statements. Briefly explain the graphical reasoning behind one of the statements. Choose one from (e)–(h). You may use 2 slides for your solution.

[1] e.g., in Section 3.3.1 of *Probabilistic Graphical Models* by Koller and Friedman.